
Session 2 (2h) — Print Settings & Common Defects

Core concepts (1h)

- **Layer height** — smaller (0.1–0.12mm) = smoother surface, much slower. Larger (0.28–0.3mm) = faster, visibly ridged. 0.2mm is the standard middle ground.
- **Infill** — the internal lattice pattern (grid, honeycomb, gyroid) and its **density** %. Higher infill = stronger + heavier + slower + more material. 15–20% is typical for prototypes that don't need to bear load; 40%+ for functional/mechanical parts.
- **Wall count / shell thickness** — outer perimeter loops. More walls = stronger and more watertight, independent of infill.
- **Print speed vs quality tradeoff** — faster speeds increase vibration artifacts (“ringing”/“ghosting”) especially on sharp corners.
- **Common defects and their usual cause:** | Defect | Usual cause | |—|—| | Warping / lifting corners | Bed too cold, poor adhesion, drafts | | Stringing (thin hairs between parts) | Retraction under-tuned, temp too high | | Layer shifting | Belt/pulley loose, print bumped | | Poor first layer | Bed not level, nozzle too far/close | | Elephant's foot (bulging bottom) | First layer squished too much |

Hands-on practice (1h)

1. Take the same test model from Session 1 and slice it three times with different layer heights (0.12mm, 0.2mm, 0.3mm) — compare estimated print time for each.
2. Slice once more at 10% infill and once at 50% infill — compare material usage and weight.
3. Using your slicer's photo/defect troubleshooting guide (most slicers or manufacturer sites have one — Prusa's and Cura's are both good references), match each defect in the table above to what it looks like in a real photo.
4. Write down, in your own words, which single setting you'd change first for each defect — this is exactly the kind of judgment call you'll make on-site when a print goes wrong.

Checkpoint

- ☐ Can explain the time/strength/quality tradeoffs of layer height and infill
- ☐ Compared time/material across at least 2 layer heights and 2 infill densities
- ☐ Can name the likely fix for warping, stringing, and a bad first layer